8.03SC Problem Set 8

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Problem 1

Consider the free transverse oscillations of the two-dimensional beaded string shown in Figure 1. The system is composed of 9 beads arranged in a 3x3 grid. All the horizontal strings have tension T_h, all the vertical strings have tension T_v, all the solid circles are beads with mass m. The ends of strings not attached to a bead are fixed. The square frame is fixed in the $z = 0$ plane.

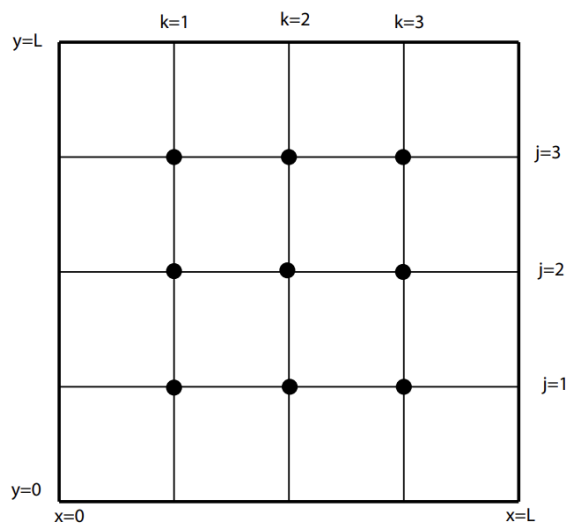

Figure 1: 2D mass lattice

a. Find the normal modes and the corresponding frequencies.

b. Suppose that $T_v = 1000T_h$. Draw nine diagrams, one for each normal mode, in order of increasing frequency, indicating which beads are moving up (by a + sign), which are moving down (by a - sign) and which are not moving (by a 0). You can interchange + and - and still have the right answer by changing the setting of your clock ir multiplying your normal mode vector by -1. For example the lowest frequency mode looks like:

+	+	+
+	+	+
+	+	+

while the mode with the fifth highest looks like:

-	0	+
0	0	0
+	0	-

a.

Considering the equation of motion, we get

$$m\ddot{z}_{k,j} = \frac{T_h}{a}(z_{k-1,j} + z_{k+1,j} - 2z_{k,j}) + \frac{T_v}{a}(z_{k,j-1} + z_{k,j+1} - 2z_{k,j}) \qquad a = \frac{L}{3}$$

Let the solution be of the form

$$z_{k,j} = \mathbf{Re}[Ae^{i(k'_x ka + k'_y ja - \omega t)}]$$

This gives us

$$-m\omega^2 = \left[\frac{2T_h}{a}(\cos k_x a - 1) + \frac{2T_v}{a}(\cos k_y a - 1)\right]$$

Therefore the Dispersion Relation is:

$$\boxed{\omega = \sqrt{\frac{2T_h}{ma}(1 - \cos k_x a) + \frac{2T_v}{ma}(1 - \cos k_y a)}}$$

Now

$$z_{k,j} = \mathbf{Re}[Ae^{i(k'_x ka + k'_y ja - \omega t)}]$$
$$\implies z_{k,j} = A\sin(k'_x ka)\sin(k'_y ja)\cos(\omega t)$$
$$\text{(-ve sign is ignored since } A \text{ is arbitrary)}$$

By the Boundary Conditions, if any of $k = 0$, $k = 4$, $j = 0$, $j = 4$ is satisfied, then $z_{k,j} = 0$. This gives us the Conditions,

$$\sin(4k'_x a) = 0 \qquad \sin(4k'_y a) = 0$$

$$\implies \quad \boxed{k'_x = \frac{n\pi}{4a}, \qquad k'_y = \frac{m\pi}{4a}, \qquad n, m = 1, 2, 3.}$$

b.

From part (a), the normal modes are

$$z_{k,j}^{(n,m)} \propto \sin\left(\frac{n\pi k}{4}\right)\sin\left(\frac{m\pi j}{4}\right) \qquad n, m = 1, 2, 3.$$

The corresponding frequencies are

$$\omega_{nm}^2 = \frac{2T_h}{ma}\left(1 - \cos\frac{n\pi}{4}\right) + \frac{2T_v}{ma}\left(1 - \cos\frac{m\pi}{4}\right)$$

Given that

$$T_v = 1000\,T_h$$

the vertical contribution dominates, hence the frequencies are primarily ordered by m and secondarily by n.

The relative motion of beads is determined by the signs of

$$\sin\left(\frac{n\pi k}{4}\right)\sin\left(\frac{m\pi j}{4}\right)$$

where $+$ denotes upward motion, $-$ downward motion, and 0 no motion.

Normal modes in order of increasing frequency:

$$(n,m) = (1,1): \quad \begin{matrix} + & + & + \\ + & + & + \\ + & + & + \end{matrix}$$

$$(n,m) = (2,1): \quad \begin{matrix} + & 0 & - \\ + & 0 & - \\ + & 0 & - \end{matrix}$$

$$(n,m) = (3,1): \quad \begin{matrix} + & - & + \\ + & - & + \\ + & - & + \end{matrix}$$

$$(n,m) = (1,2): \quad \begin{matrix} + & + & + \\ 0 & 0 & 0 \\ - & - & - \end{matrix}$$

$$(n,m) = (2,2): \quad \begin{matrix} + & 0 & - \\ 0 & 0 & 0 \\ - & 0 & + \end{matrix}$$

$$(n,m) = (3,2): \quad \begin{matrix} + & - & + \\ 0 & 0 & 0 \\ - & + & - \end{matrix}$$

$$(n,m) = (1,3): \quad \begin{matrix} + & + & + \\ - & - & - \\ + & + & + \end{matrix}$$

$$(n,m) = (2,3): \quad \begin{matrix} + & 0 & - \\ - & 0 & + \\ + & 0 & - \end{matrix}$$

$$(n,m) = (3,3): \quad \begin{matrix} + & - & + \\ - & + & - \\ + & - & + \end{matrix}$$

These nine patterns represent the normal modes ordered according to increasing frequency when $T_v \gg T_h$.